Hackathon-2

Chatbot

CHATBOT

Skill Building

What is a Chatbot?

- A Chatbot is a conversational tool to retrieve information and generate human like conversation.

- It is mainly a dialog system aimed to solve/serve a specific purpose.

Benefits and Applications

- Benefits of using a Chatbot in your Business
  - Quick information retrieval
  - Vast knowledge database
  - Can be personalized
  - Saves lots of time and money
  - Can initiate the action if programmed

- Chatbots across various Business Industries
  - Healthcare Industry
  - Educational Sector/Institution
  - Online Food Ordering
  - Customer Support
  - Banking and Finance Industry

How do Chatbots work?

Chatbots need the following to respond to an user question:

- What (skill) is the user asking for? (**Intent**)
- Did the user mention any specific information? (**Slots**)
- What should the bot ask to get further details from the user? (**Dialog flow/Maintaining Context**)
- How to fulfil the user request? (**Action/Response/Fulfilment**)

FLIGHT BOT EXAMPLE

- User Utterance: “I want to book a ticket from New York to Seattle departing on September 15 and returning on September 19”.
- Chatbot: “How many people are travelling”.
- User Utterance: “Book for 2 people”.

Utterance is the user input from which the chat bot derives intents and slots

Intent	Slots	Dialog flow	Action/Fulfillment
• Book flights	• From - New York • To - Seattle • Departure date - September 15 • Return date - September 19 • No. of people: 2 people	• Back and forth communication between bot and user	• Bot provides final results in the form of a link where reservation are made

Problem Statement

- Build an interactive conversational bot for a given Intents using 2 approaches: Alexa Chatbot and Python Chatbot.
  - Refer the Colab Notebook: Chatbot_Hackathon.ipynb

- ALEXA CHATBOT:
  - Create an Alexa Skill by logging into Alexa Developer Console
    - Refer the document: Pre-Hackathon_Alexa_Chatbot.pdf

- PYTHON CHATBOT :
  - Create a conversational Python chatbot based on the allocated Intents
    - Refer the Colab Notebook: Pre-Hackathon_Chatbot.ipynb

Alexa Skill

- Create new user account and sign in to the Alexa Developer Console. Below steps to be performed with in the Console.
  - Create two intents under one skill
  - Specify invocation name, intents, utterances and slots
  - Save and Build Model
  - Under ‘code’ tab – Build backend model by editing Lambda function
  - Test the model
  - Achieve desired outcome

Note:

- Alexa handles switching between the multiple intents.
- Use CloudWatch to debug any errors found in lambda function

Python Chatbot

- Create utterances, slots, params config file.
- Understand the different functionalities related to the chatbot, such as context, intents, slots, params and session
- Build a model to classify the intent based on utterances
- Perform the action based on the attributes
- Execute the code and run a session at the end, to test the chatbot responses

Skills to be developed

- **Zodiac Sign:** The bot should give the Zodiac Sign of the user, based on the date of birth (day, month and year) provided by the user
- **Suggest a Movie:** The bot should suggest a movie based on user preferences: Language, Actor, Genre and other details
- **Find the Restaurant:** Find the restaurants based on Cuisine, Cost type (cheap, medium, expensive), location and other parameters
- **Suggest a Book:** The bot should suggest the book based on user preferences: Author, language, genre and other parameters
- **Recommend a Store:** The bot should search a store based on preferences: Store type (medical clinics, food store, dry cleaning and more), location, availability (Open, Close) and other parameters

Intent Allocation for Teams

- Team A = Group 1,5,9,13,17,21,25=> Suggest a Movie
- Team B = Group 2,6,10,14,18,22,26 => Find a Restaurant
- Team C = Group 3,7,11,15,19,23,27 => Suggest a Book
- Team D = Group 4,8,12,16,20,24,28 => Recommend a Store

- **Zodiac Sign** is common for all teams.

- Each team allocated with 2 Intents

Criteria for evaluation of Alexa Chatbot

- **Task 1** (2 Marks) **-** Create skill and build 2 intents
- **Task 2** (4 Marks) **-** Create 50 utterances for each intent
- **Task 3** (2 Marks) **-** 3 slots and slot types for each intent
- **Task 4** (4 Marks) **-** Create a database for the intents
- **Task 5** (4 Marks) **-** Update the lambda_function.py and requirements.txt in the Code section
- **Task 6** (4 Marks) **-** Run and test the Alexa chatbot for 2 intents with the following:
  - should identify the user requirement and gather data.
  - prompt the user with different prompts and maintain dialogue flow.
  - switch between the 2 intents based on user requirement.
  - provide desired outcome based on the user requirements

Criteria for evaluation of Python Chatbot

- **Task 1** (6 Marks) **-** Create all the Utterances, Slots and update param file
- **Task 2** (2 Marks) **-** Create a database for the intent with all possible combinations
- **Task 3** (5 Marks) **-** Text representation and classification
- **Task 4** (4 Marks) **-** Define the function to perform action based on user input
- **Task 5** (3 Marks) **-** Run and test the Python chatbot for 2 intents with the following:
  - should identify the user requirement and gather data.
  - prompt the user with different prompts and maintain dialogue flow.
  - switch between the 2 intents based on user requirement.
  - provide desired outcome based on the user requirements

Evaluation & Scores

- Group activity
- One group one solution
- Hackathon will be conducted on July 18 & 19, 2026
- Presentation starts on July 19, 2026 @ 4 PM
- Scores
  - Alexa Chatbot - 20 marks
  - Python Chatbot - 20 marks
  - Presentation – 10 marks

Presentation

- 10 minutes Presentation
- Highlights
- Approaches
- Challenges Faced
- Learning Outcomes
- Teamwork

Thank you

Good Luck